



# Homecoming

A discharge-to-home copilot for family caregivers

*Nothing enters the care plan until a person confirms it.*

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SPONSOR PROBLEM

**ACL Caregiver AI Prize Challenge**

Administration for Community Living, U.S. HHS

# Discharge day is a handoff with no receiver

15–30

pages handed to a family member  
at the hospital door

*Written for clinicians. Read by a daughter  
in a parking garage.*

1

## New medications

Names, doses, and timing that did not exist last week.

2

## Changed and stopped drugs

The most dangerous line in the packet is the one that says stop.

3

## Follow-up appointments

Scattered across pages, often without dates attached.

4

## Warning signs

The symptoms that mean call now - buried in prose.

# A chatbot solves the wrong half of this

## A general caregiver chatbot

- Answers questions faster
- Generates fluent, confident text
- Explains what a term means
- Improves efficiency
- **Cannot tell you whether it made the answer up**

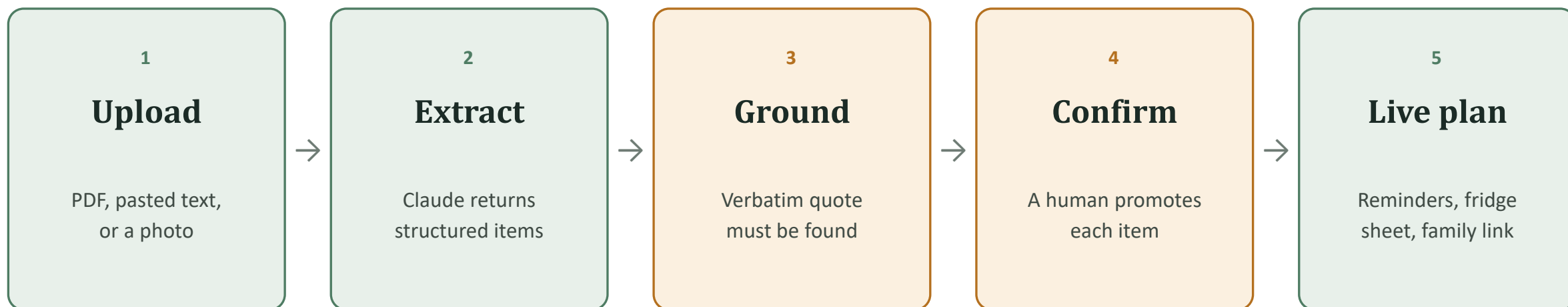
*It is also ACL's own listed use case so it is where most submissions will land.*

## Homecoming

- Takes one transition moment
- Treats it as a human-factors safety problem
- Every item traced to a page and a quote
- Prevents error, not just effort
- **Refuses to assert what it cannot ground**

*Narrow scope, verifiable output.*

# Packet in, verified plan out



**Steps 3 and 4 are the product.**

Everything else is plumbing that other teams will also have. The two gates are what make the output trustworthy enough to act on.

# Quote grounding - hallucination blocked in code

## Every item Claude returns must carry:

- a verbatim quote from the source document
- the page number it came from
- a structured payload matching a fixed schema

The function then searches the submitted page text for that exact quote.

```
extract.mjs
```

**If the quote is not found**

**the item is discarded**

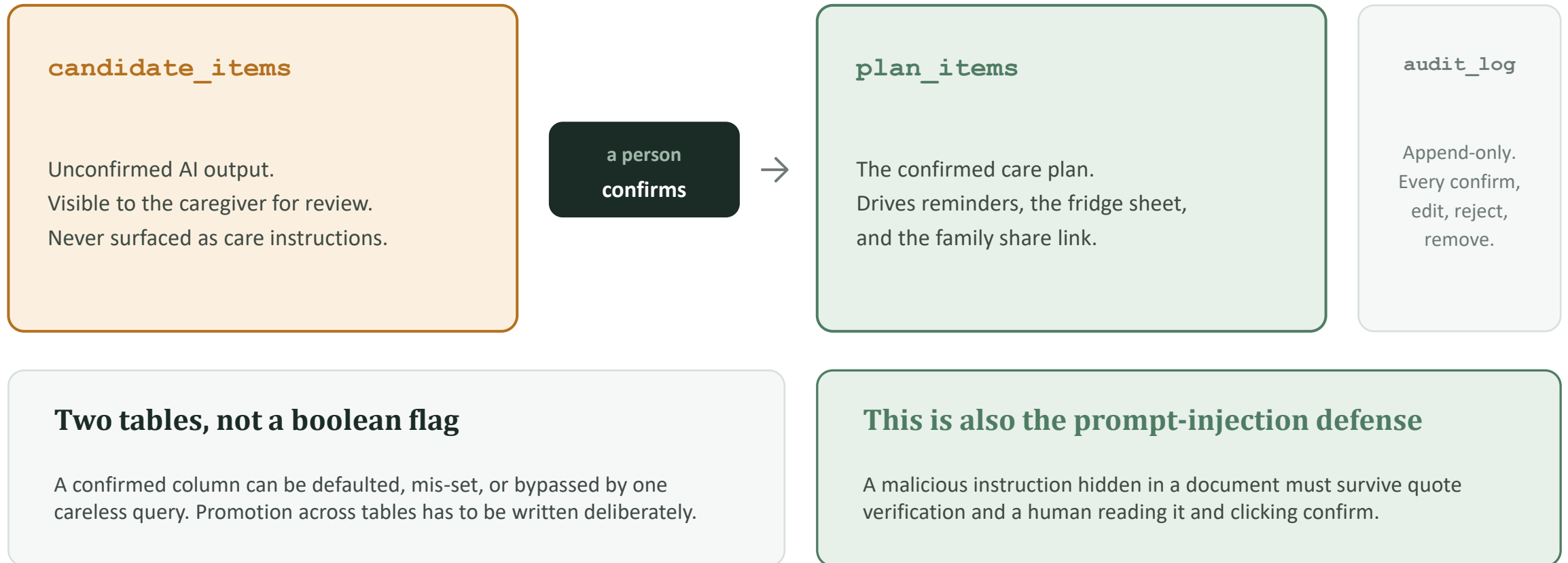
before it ever reaches the database.

***Not a warning. Not a low score. Deleted.***

## Anti-hallucination as a code path, not a prompt instruction.

Asking a model politely not to invent things is a hope. Verifying its output against the source is a guarantee.

# Human confirmation - enforced by the schema



# When there is nothing to ground against

Caregivers photograph things. A nurse's handwritten note on the back of a page is often the most important instruction in the packet - and it has no machine-readable source text to verify a quote against.

## Photos are still supported

Refusing image upload would fail the real user. Claude reads the image and extracts what it can.

## But Gate 1 cannot run

There is no submitted page text to search. The grounding check has nothing to check against.

## So they are force-flagged

Image-derived items are marked NEEDS REVIEW and can never be bulk-confirmed. Every one is read individually.

Validated against a handwritten note: the system extracted the content while correctly refusing to claim confidence in it.

LIVE DEMO

# Watch for four things

1

**The citation on every card**

Page number and source quote, on each extracted item.

2

**The amber NEEDS REVIEW flag**

Low-confidence and image-derived items, visually separated.

3

**The confirm click**

The moment an item crosses from AI output into the care plan.

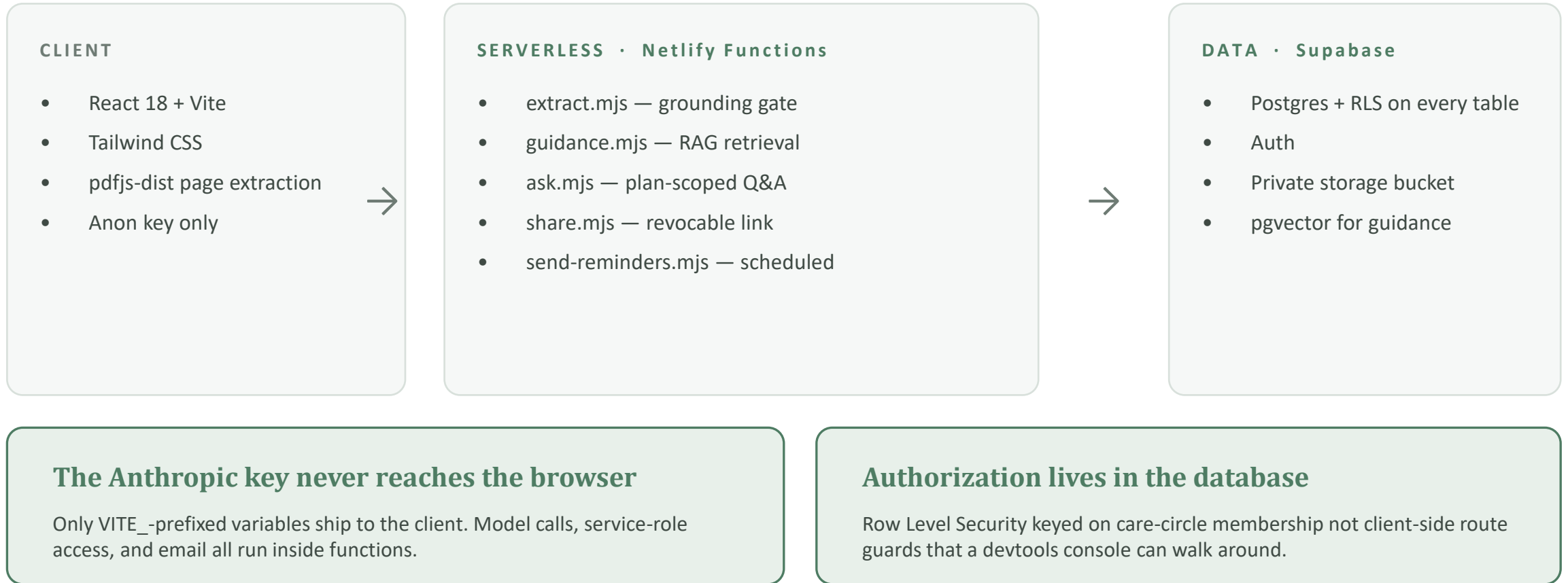
4

**The fridge sheet**

One printed page that ends up taped to a refrigerator door.



# Every secret and every check lives server-side



# Built, deployed, and running today

## ✓ Grounded extraction

PDF, pasted text, and photo ingestion with the quote-verification gate.

## ✓ Review and confirm

Confirm, edit, reject, remove — with confidence flags and an append-only audit log.

## ✓ Printable fridge sheet

The confirmed plan as one page, formatted for a refrigerator door.

## ✓ Family share link

Revocable, read-only view for relatives who are not managing care.

## ✓ RAG guidance

Answers only from a curated corpus — otherwise it declines and says ask your care team.

## ✓ FHIR export

The confirmed plan as a standards-compliant bundle, ready to hand to a clinician.

# What this system refuses to do

## No diagnosis, dosing, or triage

Claude may only restructure what the document already says. Enforced in the system prompt and validated on output.

## No ungrounded answers

If no trusted source clears the relevance threshold, guidance declines and tells the caregiver to ask their care team, rather than falling back on model knowledge.

## No real patient data, ever

All development, testing, screenshots, and demo content use one synthetic discharge summary. No real person's medical documents were used at any point.

**Homecoming organizes discharge instructions. It is not medical advice and it is not a medical device.**

# From verified plan to validated product

NEXT

## Caregiver interviews

5–8 structured interviews to trace design decisions back to lived experience — the strongest remaining gap.

IN PROGRESS

## Email reminder delivery

Schema and scheduled function are built; production delivery and deliverability testing remain.

PLANNED

## Prompt-injection test suite

Adversarial documents run as automated tests, so the injection defense is proven rather than argued.

PLANNED

## Clinical review

Bring a discharge nurse in to challenge the four item categories and the warning-sign extraction.

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**Every item in a caregiver's plan can be traced to the page it came from and a person put it there.**

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